

Review of modelling crop growth, movement of water and chemicals in relation to topsoil and subsoil compaction

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Abstract

Soil compaction influences crop growth, movement of water and chemicals in numerous ways. Mathematical modelling contributes to better understanding of the complex and variable effects. This paper reviews models for simulating topsoil and subsoil compaction effects. The need for including both topsoil and subsoil compaction results from still increasing compactive effect of vehicular pressure which penetrates more and more into the subsoil and which is very persistent. The models vary widely in their conceptual approach, degree of complexity, input parameters and output presentation. Mechanistic and deterministic models were most frequently used. To characterise soil compactness, the models use bulk density and/or penetration resistance and water content data. In most models root growth is predicted as a function of mechanical impedance and water status of soil and crop yield—from interactions of soil water and plant transpiration and assimilation. Models for predicting movement of water and chemicals are based on the Darcy/Richards one-dimensional flow equation. The effect of soil compaction is considered by changing hydraulic conductivity, water retention and root growth. The models available allow assessment of the effects of topsoil and subsoil compaction on crop yield, vertical root distribution, chemical movement and soil erosion. The performance of some models was improved by considering macro-porosity and strength discontinuity (spatial and temporal variability of material parameters). Scarcity of experimental data on the heterogeneity is a constraint in modelling the effects of soil compaction. Suitability of most models was determined under given site conditions. Few of the models (i.e. SIBIL and SIMWASER) were found to be satisfactory in modelling the effect of soil compaction on soil water dynamics and crop growth under different climate and soil conditions.

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1. Introduction

Soil compaction occurs when an applied soil stress exceeds the strength of the soil. It means by definition a reduction in porosity, and generally causes an increase in soil strength. Thereby, it changes many

properties and processes in the soil, e.g. retention, saturated and unsaturated hydraulic conductivity, air content and transport of gases, root growth and function, nutrient transport and uptake, mineralisation of nitrogen and soil workability. The still increasing vehicular pressures with their compactive effect penetrate more and more into the subsoil. This subsoil compaction proves to be very persistent. Ultimately, this will affect, e.g. crop yield, leachage, erosion and runoff, as shown schematically in Fig. 1.

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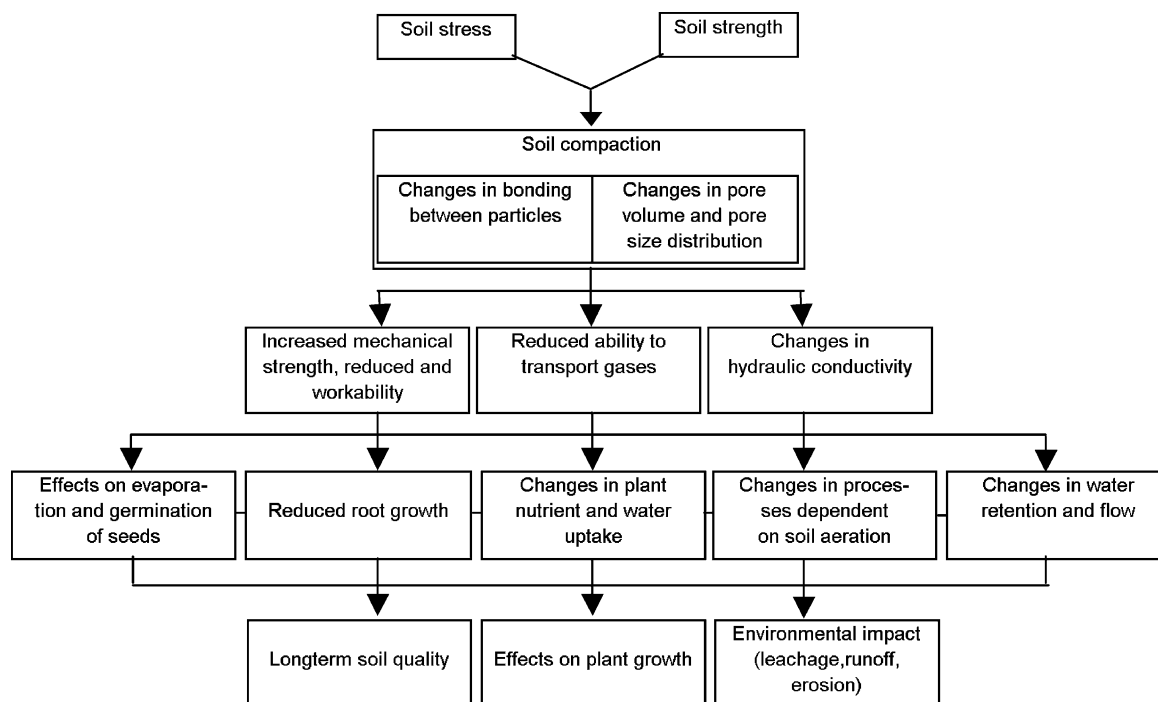


Fig. 1. Scheme showing the effect of soil stress on soil properties and processes, and ultimately on long-term soil quality, crop yield and the environment (after Arvidsson, 1997).

Relating the highly variable effects of soil compaction to crop growth and environmental responses is difficult. Mathematical modelling can contribute to understanding the complex interrelationships. Simulation models also allow us to integrate existing experimental data and to extrapolate to other site conditions. Numerous models have been constructed that deal with different parts of this process. Soil compactness is represented in the models currently available mostly by bulk density and/or penetration resistance together with water content data.

In this paper, we review models on their capability to model the effects of soil compaction on crop growth, transport of water and chemicals, and erosion. We also present opportunities for improving capabilities of the models and suggest future research needs.

2. Modelling crop growth

Over the past several years a number of crop growth models have been developed. Modelling effects of soil

structure parameters on crop growth and water balance of soil—crop systems were reviewed thoroughly (Walczak et al., 1997; Connolly, 1998). Out of 60 different crop growth models (Walczak et al., 1997) bulk density or porosity as soil compactness characteristics are represented in 32%. The most frequently represented soil characteristics are water retention (93%), unsaturated hydraulic conductivity (40%) and saturated hydraulic conductivity (33%), which are highly influenced by compaction.

The models available for predicting the effect of topsoil and subsoil compaction on crop growth vary widely in their conceptual approach, degree of complexity, output presentation and input parameter requirements. A summary of the models is presented in Table 1.

2.1. Root growth

Crop yields in compacted soils are mostly associated with the extent and function of the root system. In most models, root growth is considered as a function

Table 1
Selected models for crop growth and water and chemical movement

Model	Scale	Type	Description	Performance
SIBIL (Simota and Canarache, 1988; Simota et al., 2000)	Point	Mechanistic	Evaluates soil water dynamics and crop growth as affected by solar net radiation/air temperature/soil resistance to penetration	Performance was found to be satisfactory for simulating root growth and crop yield in compacted soil. Validated under different site conditions in Romania and Poland (Simota et al., 2000)
SIMWASER (Stenitzer, 1988)	Field	Deterministic, mechanistic	Simulates water flow and plant growth from interactions of assimilation and transpiration. Soil compaction effect is represented by penetration resistance, aeration and water content	Fairly good simulation maize yield reduction was found in field studies in Austria (Stenitzer and Murer, 2003) and Spain (Moreno et al., 2003)
Jakobsen and Dexter (1987, 1988)	Point	Deterministic, mechanistic	Simulates wheat root growth, water uptake and grain yield as affected by soil strength, macroporosity, hydraulic conductivity and water storage capacity with roots having the access to cracks in the subsoil (Jakobsen and Dexter, 1987) and to biopores (Jakobsen and Dexter, 1988)	The input soil parameters had large influences on predicted yield and were more critical under Swedish than Australian conditions (Jakobsen et al., 1989). Suitable for root growth simulation in crop growth models (e.g. Simota and Canarache, 1988; Simota et al., 2000)
Arvidsson and Håkansson (1991)	Field	Mechanistic, deterministic	Compaction effects are simulated by changing soil physical, chemical, biological and/or crop parameters. Estimates crop yield losses as affected by traffic intensity and soil water content. Effects of topsoil and subsoil compaction are treated separately	The model is based on results from northern Europe and northern America and it is not restricted to Sweden where developed
SWACRO (Feddes et al., 1984, 1988)	Field	Mechanistic	Simulates soil water movement and crop growth	Performed well in predicting subsoil compaction effects on root growth and yield of potatoes in the Netherlands
Grant (1993a,b)	Soil column	Integrated	Simulates root growth from several existing algorithms between root growth and soil strength, oxygen transport, temperature, and root turgor	Capable of simulating vertical root distribution as functions of layered soil compaction. Satisfactory evaluation was found in soil column experiment. Large computer memory required
Clausnitzer and Hopmans (1994)	Point	Mechanistic, finite-element grid	Simulates water movement and root growth direction as a function of local soil strength, temperature and texture	Permits account for the influence of soil properties on root growth and function in a single-plant scale. Good prediction of increased root density immediately above impeding layer
SOIL (Jansson, 1988)	Point	Mechanistic	Simulates water and heat flows in a layered soil with vegetation	Performed well in predicting bulk density and saturated hydraulic conductivity effects on soil water dynamics under wheat and maize in Hungary (Rajkai et al., 1997). Not satisfactory crop yield simulation in the same study
DAISY (Hansen et al., 1990)		Empirical	Simulates soil water and heat flows and nitrogen dynamics	No data

Table 1 (Continued)

Model	Scale	Type	Description	Performance
CRACK (Sławiński et al., 1996)	Field	Deterministic	Simulates bypass flow through cracks or biopores and water redistribution in the soil matrix (Green–Ampt approach) to the horizontal soil layers	Added to hydrological model ACCESS-II (Richards' one-dimensional flow) significantly improved water distribution in topsoil and subsoil (Walczak et al., 1996)
SOIL-N (Eckersten et al., 1996, 1998)	Point	Mechanistic, deterministic	Simulates nitrogen transformation and crop yield based on the availability of nitrogen. The effects of topsoil and subsoil compaction on crops can be simulated changing saturated and unsaturated hydraulic conductivity, soil water retention, root depth and distribution as input	Performed well limited N supply and maize growth in compacted soil under dry season in Hungary (Rajkai et al., 1997)
STOTRASIM (Feichtinger, 1996)	Point	Deterministic, mechanistic	Combination of SIMWASER model (Stenitzer, 1988) and a submodel to calculate nitrogen cycle dynamics. Simulates groundwater pollution by nitrate, nitrogen cycle, soil moisture regime and plant growth	Performed well leaching pattern of nitrate in comparison with data from lysimeter experiments (Dreyhaupt, 2001)
LEACHM (Hutson and Wagenet, 1992, cited by Borah and Kalita, 1999)	Point	Deterministic, mechanistic	Simulates water (Richards' eq.) and solute movement (convection-diffusion eq.) as a function of evapotranspiration and root distribution	Performed satisfactory nitrate and pesticide movement in several studies (e.g. Borah and Kalita, 1999)
RZQWM (USDA-ARS, 1992; Hanson et al., 1999)	Point	Deterministic, mechanistic	Estimates movement of nutrients and agro-chemicals in soil profile of a unit area related to soil management, bulk density and micro- and macroporosity (Green–Ampt approach)	Capable of predicting atrazine movement and concentration in soil profile. Including measured (not assumed) macro-porosity and preferential flow component improves significantly the predictions (Kumar et al., 1998). Predictions of NO ₃ -N movement were not satisfactory (Kumar et al., 1999). Large input data required
MACRO (Jarvis, 1991)	Field	Deterministic	Simulates impact of dual porosity (micropores and macro-pores) on water movement, solute transport in macroporous soils	Performed well leaching pattern of chlorsulfuron (pesticide) with the minimum calibration related to evapotranspiration, root water uptake and degradation rates in the subsoil (Bergström, 1996)
EPIC (Williams et al., 1989)	Field	Deterministic, functional	Estimates effects of soil erosion losses and cropping systems on crop growth	Use of measured values of saturated hydraulic conductivity and bulk density improved predictions of maize yield and N uptake (Warner et al., 1997a). Incorporating a submodel of root growth as a function of bulk density affected simulation of root growth of a dryland cotton (Bland and Jones, 1992)

of soil penetrometer resistance and water status. This effect can be estimated by an impedance function that is unity at zero soil strength and decreases linearly to zero at a critical soil strength value at which root growth stops (Dexter, 1987; Diggle, 1988; Bengough and Mullins, 1990; Bland and Jones, 1992). Root growth as affected by soil water matric potential and resistance can be simulated by the SIBIL model (Simota et al., 2000) in which the theory of root growth mechanics developed by Dexter (1987) is used. Soil resistance is quantified in the model using a semi-empirical model (Canarache, 1990) with clay content, bulk density and gravimetric water content as input parameters. The model SIBIL was a good predictor of vertical distribution of root mass in variously compacted soil (Fig. 2).

The ability of roots to penetrate strong soil can increase with increasing root hardness (Panayiotopoulos et al., 1994). Stenitzer (1988) assumed in the SIMWASER model (Table 1) that depending on 'root type' the values of penetrometer resistance that cause beginning reduction in root growth vary from 1 MPa ('low root strength') to 1.7 MPa ('high root strength') while those stopping root growth vary from 3 to 4 MPa. The critical strengths may vary depending on soil texture, macroporosity, depth and crop type (Gliński and Lipiec, 1990; Pabın et al., 1998).

It was shown that the values of critical soil strength and bulk density for pea root growth in the subsoil decreased with increasing content of fraction <60 μm and with decreasing soil water content (Fig. 3). However, in plough layers an increase in the content of this fraction resulted in greater critical soil strength for root

growth due to the favourable effect of organic carbon content. The presence of pores greater than roots in dense layers tends to move the critical soil strength to a higher value (Ehlers et al., 1983). Setting appropriate values is thus of great importance for good simulation (Stenitzer and Murer, 2003; Moreno et al., 2003). Scarcity of experimental data on the 'realistic penetration resistance' is the major constraint on predictive capacity of the models (Simota et al., 2000; Stenitzer and Murer, 2003).

Soil aeration compared to soil strength was less frequently represented in crop growth models (Walczak et al., 1997). The review by Gupta and Raper (1994) indicates that an approach assuming optimum root growth in the range of soil wetness between 0 and 85% of saturation and linearly decreasing root growth from 85 to 100% saturation is often used. Thus a negative effect of insufficient aeration in compacted soil will increase in soils prone to wetness (Boone et al., 1986). Insufficient aeration is often short lasting and difficult to relate to crop yield response due to soil compaction (Stępniewski et al., 1994; Håkansson and Lipiec, 2000). As indicated by Van Huyssteen and Ellis (1997) the average duration (percentage of the monitoring period) of saturation with water was 1.3% for red apedal B horizons, 18.8% for yellow-brown apedal B horizons, 42.4% for yellow E and 54.2% for grey E horizons.

2.1.1. Effects of soil structural discontinuities

An important factor influencing predictability of root profile distribution is strength discontinuity. It may occur between the aggregated seedbed and the

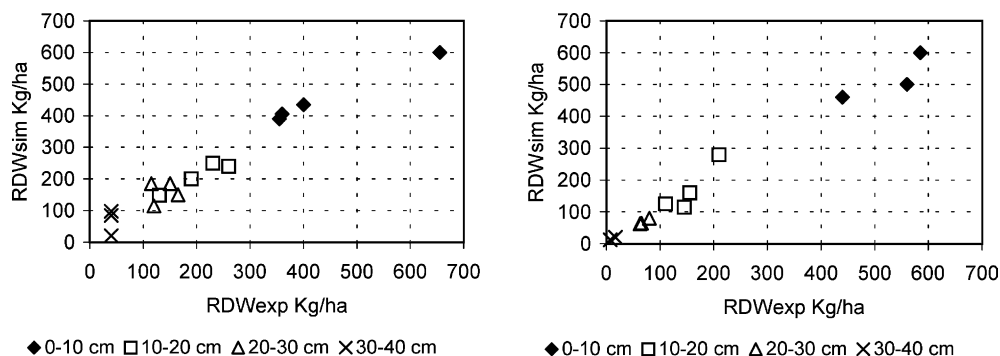


Fig. 2. Simulated vs. observed root dry weight of spring barley (at heading) for non-compacted (left) and compacted (right) soil (after Simota et al., 2000).

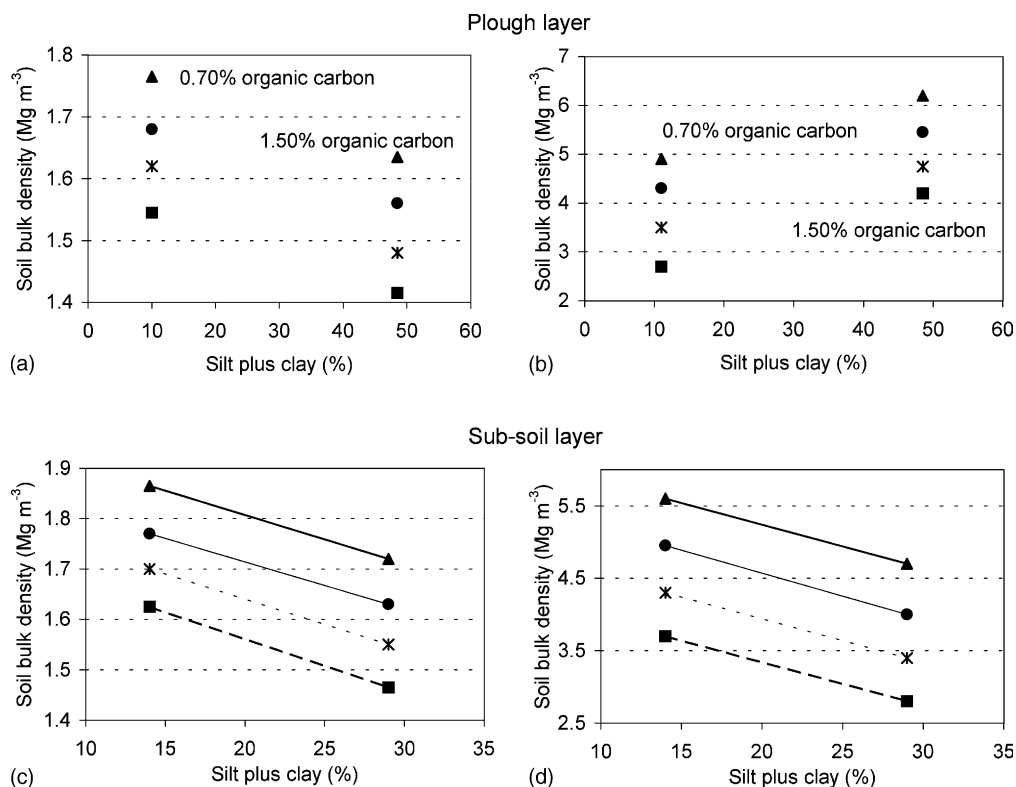


Fig. 3. Critical soil bulk density for root growth in relation to silt plus clay content, organic carbon and water level in the plough and subsoil layer. (■) 30; (✱) 40; (●) 50; (▲) 60% field water capacity (after Pabin et al., 1998).

soil below after seedbed preparation, between a tilled layer and untilled subsoil, or across soil horizons and various soil structural units (Dexter, 1986; Smucker and Aiken, 1992; Lipiec et al., 1993). Table 1 presents some of the models for simulating root growth in structurally discontinuous soils. Several models performed well in predicting a higher concentration of roots in the top layer of compacted soil and decreased rooting depth in field experiments (Simota, 1993; Simota et al., 2000) or in a soil column (Grant, 1993b). Using discretised representation of a finite soil domain and local soil strength gradient in a study by Clausnitzer and Hopmans (1994) allowed calculation of root growth direction vector and prediction of increased root density in a single-plant scale immediately above impeding layer. The presence of the structural discontinuities (dense layers) in soil profile limits application of root growth models assuming exponential declining root growth with depth (Gerwitz and Page, 1974; Feddes

et al., 1988) and those using elastic modulus (ratio of applied pressure and resulting deformation) as a soil strength parameter controlling root growth (Rickman et al., 1992; Grant, 1993a,b).

The discontinuities have a significant effect on root absorption. Subsoil layers are usually wetter and provide greater root–soil contact and consequently can be a more effective medium for transmission and uptake of water by roots. It was shown in growth chamber experiments that root length of maize in subsoil horizons (30–50 and 50–70 cm) relative to total root length varied from 1 to 38% while water use varied from 54 to 74% depending on soil type (Lipiec et al., 1993). As a consequence root water uptake is not always linearly correlated with greater root size. Some models allow for a preferential uptake from wet compared to dry sites (De Willigen and Van Noordwijk, 1987) and anoxic sites (Schmidhalter et al., 1994).

The profile discontinuities make it difficult to separate the effects of topsoil and subsoil compaction (Lindstrom and Voorhees, 1994). Smucker and Aiken (1992) in their review indicated that to describe root functions successfully we need to move away from the simplistic approach that water is absorbed uniformly in the soil profile and to include enhancements of the Darcy/Richards equation to predict the movement of water across structural discontinuities. In addition incorporating into models compensatory root growth and uptake in favourable local environments and short time intervals (diurnal or hourly) of the dynamic root and soil interface may increase performance of the models. Clothier and Green (1997) indicate that Time Domain Reflectometry for measuring soil water content close to roots and near the soil surface in short time steps provides means by which we can better view the root zone fluxes of water and chemicals.

Another structural discontinuity is the presence of macro-pores of a diameter greater than the roots. The significant importance of continuous and stable macro-pores such as inter-pedal voids, bio-pores or desiccation cracks for root growth as well as for gas, water and solute transport has been reported by several authors (Hatano and Sakuma, 1990; Whalley and Dexter, 1994; Lipiec and Stępniewski, 1995). The continuous bio-pores may be particularly advantageous in strong subsoil horizons because they frequently provide the only possible pathways for root growth and are resistant to vertical compression (Ehlers et al., 1983; Whalley and Dexter, 1994).

The relationship between the distribution of macro-pores and roots can be described numerically using fractal analysis (Hatano and Sakuma, 1990; Lipiec et al., 1998). The studies showed a relationship between fractal dimensions of distribution patterns of macro-pores and plant roots in strong soil. Macro-pores significantly influenced root growth in modelling work by Jakobsen and Dexter (1987, 1988) where roots entering the pores could avoid mechanical impedance.

2.2. Crop yield

There are several models available for predicting the effects of topsoil and subsoil compaction on crop growth and yield (Feddes et al., 1984; Stenitzer, 1988; Simota et al., 2000) (Table 1). They are mostly de-

signed for simulating water flow and dynamics using the Darcy/Richards one-dimensional flow equation for water and heat flow based on Fourier's law (Jansson, 1988). Unsaturated hydraulic conductivity is calculated from the water retention curve and is matched at saturated conductivity. Water balance and plant growth can be linked together by physiological interactions. The actual plant growth is calculated by the potential production rate as the proportion of actual transpiration to potential transpiration. Soil compaction effects can be simulated directly by considering bulk density or porosity (Walczak et al., 1997) or indirectly by changing inputs for saturated hydraulic conductivity, soil water retention, unsaturated hydraulic conductivity, penetrometer resistance, root depth and root distribution (Rajkai et al., 1997; Warner et al., 1997a,b; Eckersten et al., 1996, 1998).

Arvidsson and Håkansson (1991) constructed an empirical model to predict effects of compaction on crop yield based on results from a large number of experiments, mainly carried out in Sweden. Arvidsson et al. (2000) used the model to calculate expected yield loss due to sugarbeet harvest, with a selfpropelled six-row harvester and a three-row harvester towed by a tractor. They expected similar losses due to compaction of the topsoil for the two systems, but higher yield loss due to subsoil compaction for the selfpropelled harvester because of higher axle load. To adjust the model to other regions, local yield data is of great value. Jakobsen and Dexter (1988) constructed one of the few models that simulate soil water flow and crop growth, and where the impact of soil strength on root growth can be calculated.

3. Modelling of water and movement of chemicals

Most models for simulating movement of water and chemicals are based on the Darcy/Richards one-dimensional flow equation (Walczak et al., 1997; Connolly, 1998). The effects of soil compaction on water movement and redistribution in the soil profile is mostly through changes in hydraulic properties (Walczak et al., 1997) and indirectly through influences on soil mechanical resistance and aeration status and related root growth and uptake (Horton et al., 1994). Numerical models for prediction of field

compaction effects on soil water and thermal regimes were reviewed by Horton et al. (1994). The authors indicate that simulated water and heat flow are sensitive to traffic compaction and ridge configuration. Wu et al. (1999) reported that the presence of a strong layer in the subsoil (30–40 cm) of sandy soil retarded downward water movement and as a consequence the water contents in deeper soil, as predicted by root zone water quality model (RZQWM) (USDA-ARS, 1992; Hanson et al., 1999), were greater than measured ones. The authors indicate that the capability of the model can be improved by incorporating physical and hydraulic parameters of the impeding subsoil layers as adequate input. An appropriate prediction of soil water movement is required for accurate modelling of solute transport (Jarvis, 1991; Feyen et al., 1998). Chemical leaching from agricultural fields is a major source of contamination for water resources in many regions. The effects of soil compaction on components of the environment are illustrated in Fig. 4.

The effects of soil compaction on soil nitrogen dynamics can be simulated by the STOTRASIM model (Feichtinger, 1996) (Table 1). The main attention is

directed to the environmental aspect of groundwater pollution by nitrate. STOTRASIM also calculates soil moisture regime and plant growth and the nitrogen cycle for soils used for agriculture. For this, precipitation, irrigation, fertilisation, evaporation and interception, plant uptake, volatilisation, denitrification, mineralisation, immobilisation and the storage and transport in the soil are considered. Castle et al. (1999) showed that the potential for denitrification in glacial till subsoil layers is sufficient to reduce NO_3^- leaching to ground- or surface-waters to levels unlikely to result in pollution hazard. The major product of NO_3^- reduction in these subsoil layers was N_2 , rather than the greenhouse gas and catalyst of stratospheric O_3 removal, N_2O . Warner et al. (1997a) reported that nitrate concentrations in a fine sand loam underlain by a compact glacial till can be predicted by the EPIC model. The predictions were better for fallow than maize and for deeper than shallow soil depths. Concentrations of nitrate in soil profile were over-predicted by the RZQWM model (Kumar et al., 1999) likely due to using only bulk density as an input parameter of soil physical state. Model predictions of nitrate movement are very sensitive to air-filled porosity and oxygen

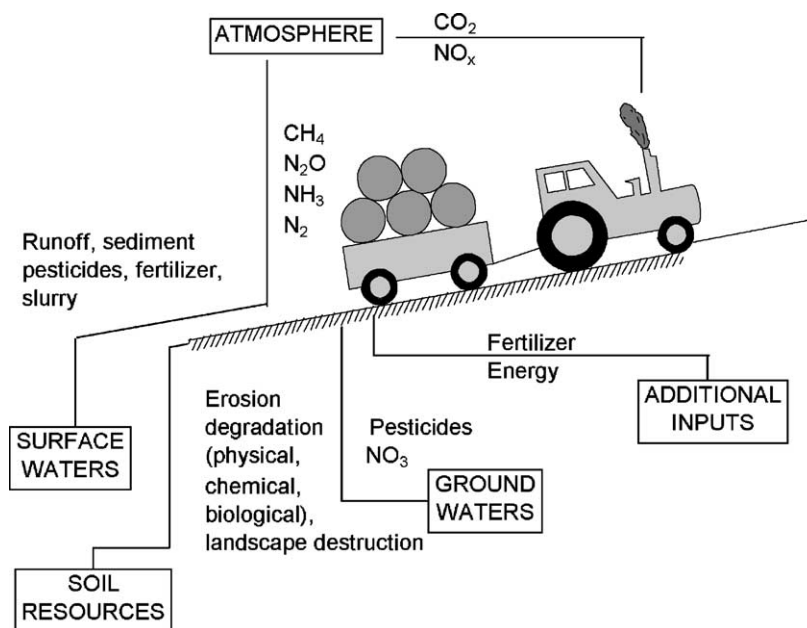


Fig. 4. A conceptual diagram showing the influence of soil compaction on components of the environment (after Soane and Van Ouwerkerk, 1995).

diffusion parameters through their strong influence on transformation processes of mineralisation, nitrification and denitrification (Hendriks et al., 1999).

Many soil nitrogen models were designed for advising farmers and their suitability for estimating environmental impacts is limited (O'Sullivan and Simota, 1995). Models for simulating agricultural chemical transport in lysimeter conditions were reviewed by Borah and Kalita (1999).

3.1. Effects of macro-pores

A soil matrix with macro-pores such as inter-pedal voids, bio-pores or dessication cracks offers greater potential for undisturbed growth of roots and has a significant effect on the water flow and solute transport processes. Compaction has a great influence on macro-pore flow, but there have been few attempts to model these effects. Under a flow at 0 cm tension, macro-pores >0.5 mm and meso-pores 0.06–0.5 mm (radius for cylindrical pores or width for planar pores) contributed about 89 and 10% of the total water flux, respectively (Lin et al., 1996). The presence of macro-pores is distinguished as soil micro-heterogeneity (at the pore scale) (Feyen et al., 1998). It was shown that incorporating the flow macro-pore component into models that assume a horizontally homogenous soil profile improves their performance (Jarvis, 1991; Ludwig et al., 1999; Borah and Kalita, 1999; Guber, 2000). A close negative relationship exists between soil compaction and total volume of macro-pores (Carter, 1990; Diserens et al., 1998; Lipiec and Håkansson, 2000) and that of pores stained with dye in macro-pore flow experiments (=macro-pores actively contributing to water flow) (Lipiec and Håkansson, 2000). Models to estimate the effect of tillage and consolidation on the water retention curve, hydraulic conductivity and pesticide leaching were presented by Ahuja et al. (1998) and Rojas and Ahuja (1999). Zhuang et al. (2000) also modelled the effect of compaction on saturated hydraulic conductivity. In practice, there is no simple relationship between bulk density and saturated hydraulic conductivity. In reduced tillage, saturated hydraulic conductivity has often been found to increase despite of a higher bulk density, while in other experiments a small increase in bulk density may decrease the conductivity by several orders of magnitude (Arvidsson, 1997).

Sławiński et al. (1996) developed the CRACK sub-model of bypass flow through cracks or macro-pores and water redistribution in the soil matrix following infiltration to the horizontal soil layers using the Green–Ampt approach. The sub-model enhanced performance of the ACCESS-II hydrological model (based on Richards' one-dimensional flow) in predicting water distribution in plough and subsoil layers in two sites in Poland (Walczak et al., 1996). By replacing the air-filled porosity in the Green–Ampt model with the active macro- plus mesopores, calculated wetting front depths were close to observed maximum dye depth (Lin et al., 1996). Simultaneous ground and macro-pore flow occurs in all except very coarse soils (Hatano et al., 1992; Horton et al., 1994; Feyen et al., 1998).

Macro-pore flow significantly influences pesticide movement in soil. Sadeghi and Isensee (1992) reported that atrazine movement in the subsoil was more affected by the presence of macro-pores in the no-till than in conventionally tilled plots. Atrazine residues in the topsoil layer were higher in conventionally tilled than in no-till soil due to the presence of a plough pan at about 30 cm depth. The movement of atrazine through macro-pores is greatest when high-intensity rainfall occurs shortly after atrazine application (Edwards et al., 1993). It was shown that prediction of pesticide leaching by the deterministic model MACRO (Jarvis, 1991) is highly sensitive to soil hydraulic properties defining the macro-pore region (e.g. hydraulic conductivity at the boundary water content). The model successfully described the leaching pattern of chlorsulfuron (pesticide) with the minimum calibration related to evapotranspiration, water uptake by roots and degradation rates in the subsoil (Bergström, 1996).

Several studies have revealed that incorporating the macro-pore flow subroutine improved prediction of atrazine concentration by the one-dimensional Richards flow models leaching estimation and chemistry model (LEACHM) (Borah and Kalita, 1999) and RZQWM (Kumar et al., 1998). However, there was not significant effect of macro-pore flow in the LEACHM on simulations of NO₃–N transport which can be attributed to high sensitivity of the model to nitrogen and carbon pools in the soil profile (Borah and Kalita, 1999). Hatano et al. (1992) using fractal analysis found that macro-pore flow contributes

significantly to patterns of miscible displacement and is different depending on the type of macro-pores and the smoothness of their walls. Hendriks et al. (1999) indicated that considering both dynamic shrinkage cracks and permanent macro-pores allow for better model simulations of solute transport.

The effect of macro-pore on transport of nutrients increases with their initial concentrations in soil. These were considered in prediction of phosphate (Hansen et al., 1999) and nitrate leaching (Jarvis, 1991).

Movement of chemicals in macro-pores depends on sorption capacity of the soil. Several authors (Levanon et al., 1993; Cox et al., 1998; Hansen et al., 1999) reported that lower leaching of agro-chemicals in macro-porous soil can be attributed to enhanced sorption (in no-till due to greater organic matter and lower pH).

An empirical model study on four macro-pore soil materials in the Eg and Btg horizons showed that the risk of phosphate leaching through subsoil macro-pores is significantly affected by P sorption capacity of the soil materials closest to the lumen of macro-pores (Hansen et al., 1999). Macro-pore walls in fractures (but not in earthworm burrows) in the albic material (poor in iron oxide and bleached) with the smallest P adsorption capacity do not minimise the risk of phosphate leaching. However, usually greater adsorption is predicted when an average bulk sample is considered. Therefore, sorption properties of macro-pore materials need to be included in models describing transport of reactive substances in macro-porous subsoil layers. Because of fast water flow in macro-pores, sorption reactions are far from equilibrium and the modelling studies should consider a fast initial reaction (time scale of milliseconds to seconds). Preferential transport of phosphate is of great importance in the clayey Pseudogley soils with low adsorptive subsoil layers and phosphate rich plough layers. Lower leaching of agrochemicals in macro-pores can also be reduced by degradation.

Inclusion of measured compared to estimated macro-porosity (assumed to be reduced proportionally to an increase in bulk density) enhanced model performance in movement and concentration of chemicals (Borah and Kalita, 1999; Kumar et al., 1999). In cracked clay soils the shrinkage characteristic is used in modelling to determine crack volume, area and depth and leaching by fast moving water through

cracks from topsoil to subsoil (Kroes et al., 1998, 2000).

The recently developed numerical model soil–water–atmosphere–plant (SWAP) (Kroes et al., 1998, 2000) is the successor of the agro-hydrological model soil water-actual transpiration (SWATR) (Feddes et al., 1978) and some of its numerous derivatives. The main improvements are accurate numerical solution of the Richards flow equation and incorporation of solute transport, heat flow, soil heterogeneity, drainage and water management. It integrates water flow, solute transport and crop growth according to current modelling concepts and simulation techniques. The model performed well in predicting soil moisture content in direct drilled soil where the time-dependency of the soil hydrophysical functions did not influence strongly simulated water balance (Farkas et al., 2000) and can be useful in predicting the effects of soil compaction.

4. Soil erosion

Effects of soil compaction on erosion may have a serious environmental impact. The process may occur due to a decrease in the infiltration rate of the soil, e.g. in wheel ruts, or due to layers with low hydraulic conductivity, such as plough pans. Fleige and Horn (2000) measured a great increase in erosion in wheel ruts compared to untrafficked land. Bazoffi and Pellegrini (2000) found that wide low-pressure tyres caused a significant increase in erosion compared to normal tyres, possibly because the wheel tracks of the wider tyres covered a larger part of the field.

Numerous models exist to simulate erosion under various conditions. One of the most widely used is revised universal soil loss equation (RUSLE) (Renard et al., 1991), which is mainly an empirical model based on properties of the soil surface and the crop canopy. Therefore, it is not really suitable for simulating effects of subsoil compaction on erosion. For this purpose, more physically based models such as WEPP (Laflen et al., 1997) and LISEM (De Roo and Wesseling, 1996) are more suitable. LISEM is a physically based single-event hydrological and soil erosion model. For example, it can take into account the hydraulic conductivity of different soil layers, and the effect of wheel tracks on the soil surface. WEPP is also a physically based model to simulate runoff and erosion.

It makes daily simulations based on soil and crop properties, driven by meteorological data and should be suitable for simulating effects of subsoil compaction on erosion.

5. Spatial variability in modelling of soil parameters

Spatial variations involved with the input parameters may reduce the certainty of the model output data (Horton et al., 1994; Verma et al., 1995). The spatial distribution within the field is closely related to the distribution of wheel tracks (Arvidsson and Håkansson, 1991; Walczyk, 1995). Knowledge of the spatial dependence of penetration resistance may reduce the number of penetrations and sampling positions for accurate evaluation of the efficiency of tillage practices and mechanical impedance for root growth. Geostatistical analysis showed that the range of influence (the distance over which the semi-variance increases) was greater in compacted than loose soil (Lipiec and Usowicz, 1997). In non-compacted soil, the range was greater in stronger subsoil than in the plough layer. The results imply that sampling interval for representative results should be smaller in loose soil. Chan and Hodgson (1995) reported that strength of the subsoil of a Vertisol was significantly higher and less spatially variable in a cropped soil compared with a pasture soil, indicating loss in the heterogeneity and changes in structural organisation as a result of cropping. Greater variability of pore distribution patterns in loose than compacted soil was shown by fractal analysis (Lipiec et al., 1998).

Koszinski et al. (1995) reported that clay and total C contents, dry bulk density and rootability in the subsoil (at 55 cm) were autocorrelated over a distance of about 25–50 m, whereas soil structural parameters, such as numbers and area of macro-pores as well as permeability properties, varied randomly even over the shortest distance (10 cm). Both in the topsoil (15 cm) and plough pan (30 cm) most parameters show random variation over short sampling distances of 10 cm. This information is of significant importance for spatial interpolation of soil properties.

Spatial variability in soil hydraulic properties and water significantly influences model predictions of main water budget components such as evaporation

and drainage storage (Maraux et al., 1998). Simulation agreement with measured data was the highest in the less variable subsoil layer 60–100 cm. Nitrate concentration was also better predicted at deep compared to shallow depths using the EPIC model in a fine sandy loam underlain by compact glacial till (Warner et al., 1997a,b). Spatial variability of the hydraulic functions in models can be described with the scaling concept. In the SWAP model (Kroes et al., 1998), the reference hydraulic functions and the corresponding water and solute balance and relative crop yield are generated by a number of scaling factors. Pachepsky et al. (1995) showed that scaling of soil water retention can be described by a fractal model. The scaling is an important issue in dealing with a quantitative prediction at field or regional scales (Feyen et al., 1998). Soane and Van Ouwerkerk (1995) stress that the spatial variability of soil properties, both vertically and horizontally, should be taken into account in studies of soil compaction effects.

6. Conclusion

Soil compaction effects on crop growth and water and chemical movement can be predicted by deterministic, mechanistic, empirical and finite-element models depending on the aim of simulation and applicability. Soil compactness is mostly represented in the models directly by bulk density (or total porosity), penetration resistance and water content or indirectly by water retention and hydraulic conductivity. The model parameter values vary depending on soil texture, depth, crop and root type. There is a considerable potential to improve model performances by incorporating the relationships between macro-pores and root growth, water and solute flow and sorption characteristics. The model prediction is sensitive to spatial and temporal variability in input parameters that are largely influenced by compaction. Several model parameters exhibited lower spatial variability in the subsoil than in the topsoil. Further work is needed to develop modelling approaches to describe more accurately the effects of soil structural discontinuities (i.e. strength discontinuity, macro-pores) and spatial variation of input parameters resulting from compaction.

Compaction models derived from soil mechanics generally aims at predicting changes in bulk density

but not in other soil properties, such as saturated and unsaturated hydraulic conductivity. These are important parameters in most models where soil physical processes are simulated. It is important to overcome this gap between models to predict the final effects of compaction, in terms of crop growth and environmental effects, such as erosion and nitrogen and pesticide leaching.

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